

Canadian Donations Mapping Platform (CDMP)

1. Executive Summary

The fundamental problem that CDMP aims to address is the prohibitive difficulty of exploring Canadian federal political-donation data in its current form. At present, anyone wishing to understand the geographic distribution of political contributions must locate and download numerous raw data files published by Elections Canada, manually reconcile inconsistent formatting across reporting periods, crosswalk postal codes to electoral-district geographies, and then produce their own visualizations from scratch. For a trained research assistant, this process takes several weeks; for a journalist or undergraduate student, it is effectively inaccessible.

CDMP is intended to compress that entire workflow into a task that takes less than one minute. Specifically, the envisioned product is an interactive, map-centred web application that allows users to select a geographic level of analysis, filter by time period and political party, and immediately view a choropleth map shaded by donation volume. Users should be able to click on any region to retrieve summary statistics, party-level breakdowns, and longitudinal trend information.

The project carries both a public-facing dimension and a research dimension. On the public side, the application should serve aggregated, anonymized data to any visitor, thereby opening up a dataset that is technically public but practically opaque. On the research side, authenticated university-affiliated users should be able to access individual-level donation records through a secured login, subject to appropriate privacy protections.

2. Background

One of the Policy Horizons Lab's (UTSC) ongoing lines of inquiry concerns the spatial distribution of political finance in Canada. Elections Canada publishes individual-level contribution records for every federal political entity — parties, electoral district associations, leadership contestants, and nomination contestants — and has done so, in varying formats, for roughly two decades. In principle, this data allows any member of the public to see who is giving money, how much they are giving, to which party, and from which postal code. In practice, the data is fragmented across dozens of separate CSV files, each structured slightly differently, with postal codes that frequently contain typographical errors and with no direct linkage to standard electoral geographies.

The research group has invested considerable effort in building internal data pipelines that clean, geocode, and merge these records into analytically useful datasets. However, these pipelines are research-grade scripts — they require programming expertise to run, they lack a user-facing interface, and they are not suitable for distribution to a broader audience.

3. Problem Statement

The core problem can be stated as follows: a dataset that is, in a legal sense, already public is, in a practical sense, almost entirely inaccessible to the constituencies that would benefit most from examining it. Journalists covering campaign-finance stories cannot quickly check whether a particular riding saw

unusual donation patterns. Political-science undergraduates writing term papers cannot easily compare per-capita giving across provinces. Civic organizations interested in democratic transparency cannot offer their members a straightforward way to see where political money comes from.

The barriers are not primarily legal or administrative; they are technical. The raw data requires substantial preprocessing before it yields meaningful insight, and the geographic dimension of the data — arguably its most interesting feature — is entirely absent from the flat files that Elections Canada publishes. Bridging the gap between the raw data and a meaningful visualization requires skill in data cleaning, postal-code geocoding, geospatial file manipulation, and web-application development. No single tool currently provides this bridge.

CDMP is intended to fill this gap. The application should accept the cleaned, geocoded data that the research group already produces, store it in a performant database, expose it through a backend API, and render it in a browser-based frontend that foregrounds geographic exploration.

4. Intended User Populations

The application is designed to serve two distinct user populations, each with different access levels and different analytical needs.

Public Users

The first population consists of members of the general public, including journalists, students, civic-engagement organizations, and casual browsers. These users should be able to access the application without creating an account or providing any credentials. They will interact with aggregated data only: total donations by region, party, and year, with appropriate suppression of small cells to prevent the re-identification of individual donors. The interface presented to public users should be intuitive enough that a journalism student with no technical training can produce a meaningful map within sixty seconds of arriving at the site.

Authenticated Research Users

The second population consists of university-affiliated researchers — primarily graduate students and research assistants working under the supervision of group members. These users require access to individual-level donation records, including donor names, exact amounts, and precise postal codes. Because this level of detail raises legitimate privacy concerns, access to the research tier must be gated behind an authentication mechanism. All research-tier access should be logged so that the group can maintain an audit trail of who accessed individual-level data and when.

5. Data Environment

Category	Format	Characteristics
Donation records	CSV	Millions of rows spanning roughly twenty years. Each record contains a donor identifier, contribution amount, date, recipient political entity, and a postal code. Postal codes frequently contain typographical errors that must be handled during ingestion.

Electoral-district boundaries	Shapefile / GeoJSON	Polygon geometries representing the boundaries of federal electoral districts. Boundaries have been redrawn three times during the period covered by the data, so the system must store and distinguish among multiple boundary vintages.
Postal-code lookup table	CSV	Maps each six-character Canadian postal code to a latitude–longitude pair and a federal electoral district code. This table is the critical linkage between the donation records and the geographic boundaries.
Census population data	CSV	Population counts at the riding and FSA level, used to normalize donation totals into per-capita metrics. Without this normalization, high-population urban ridings would always dominate the map.
Pre-rendered map tiles	Raster tiles or vector tiles	Canada’s geographic extent is large, and rendering full polygon layers in the browser at national scale will degrade performance. A tiling strategy is therefore required.